[I225] Statistical Signal Processing(E) Office Hour 1

1. Imagine you are looking at the digital thermometer outside the university. We want to model the temperature recorded there.

   Let $T(t, \omega)$ be the stochastic process of temperature at time t (hours past midnight).
   The randomness ω comes from which specific **day** of the year we are looking at (e.g., April 1st vs. October 1st).

   A. Classify the following events: (constant value, random variable, stochastic process, or realization)
      [1] The predicted temperature curve for tomorrow, from 0:00 to 23:59.
      [2] The temperature measured at exactly 8:00 AM across every possible day of the year.
      [3] The collection of every temperature reading ever taken at this station at every second of every day.
      [4] The temperature at exactly 12:00 PM on January 1st, 2024.
      [5] A value representing the average yearly temperature of the city.

   B. Based on how temperature behaves throughout a 24-hour cycle, would you classify the temperature process $T(t, \omega)$ as Wide-Sense Stationary (WSS), Strict-Sense Stationary (SSS), or Non-Stationary? Justify your answer using the **average** of the process.

   C. Suppose you want to find the average temperature of the city. You have two choices:
      a. **Time Average:** Record the temperature of **one specific day** and average it over 24 hours.
      b. **Ensemble Average:** Measure the temperature at **exactly 12:00 PM** across 365 different days and average those values.

      Will these two averages give you the same result? Based on this, is the process **Mean-Ergodic**?

A. Definition of Stochastic Process

Item	Classification	Conceptual Explanation
1	Realization	It is a single function of time. Once the "day" is picked, there is no more randomness; it's just a line on a graph.
2	Random Variable	Time is "frozen". We are looking at a single moment, but the value changes depending on which day you pick.
3	Stochastic Process	This is the "Whole Thing". It includes all the days (randomness) and all the hours (time).
4	Constant	Both time and the day are fixed. It is a single, historical measurement.
5	Constant	It's just a number. It doesn't change with time or which day you look at.

B. Stationary

Classification: Non-Stationary.

Reasoning:
For a process to be WSS (and by extension SSS), the **mean** must be constant for all time: $E[T(t)] = \eta_T$.

In our weather model, the ensemble mean $E[T(t)]$ represents the average temperature at a specific hour t across many days. Because of the solar cycle, the average temperature at 2:00 PM is significantly higher than at 2:00 AM. Since $E[T(14{:}00)] \neq E[T(02{:}00)]$, the mean is **time-dependent**, making the process **non-stationary**.

C. Ergodicity

Classification: Not Ergodic.

Reasoning: A process is Mean-Ergodic if the **time average** of a single realization converges to the **ensemble average**.

- The **time average** of one day captures the full cycle (cold night to warm day).
- The **ensemble average** at 12:00 PM only captures the "peak" heat of the days.

Since the average of a single "slice" of time (the whole day) does not represent the statistical average of the "whole group" at a fixed moment (noon), the process is not ergodic.

2. Consider a signal that has a constant DC offset A plus a random oscillating component:

$$X(t) = A + \cos(\omega_0 t + \Theta)$$

- A is a constant.
- Θ is a random variable uniformly distributed between $[0,2\pi]$.

A. Calculate the average: $E[X(t)]$

B. Calculate the Autocorrelation: $R_{XX}(t_1, t_2)$

C. Calculate Autocovariance: $C_{XX}(t_1, t_2)$

A. Calculate the average: $E[X(t)]$

$$E[X(t)] = E[A + \cos(\omega_0 t + \Theta)]$$

$$= E[A] + E[\cos(\omega_0 t + \Theta)]$$

$$= A + \int_{-\infty}^{\infty} \cos(\omega_0 t + \Theta) . f(\Theta)\, d\Theta$$

$$= A + \int_{0}^{2\pi} \cos(\omega_0 t + \Theta) \frac{1}{2\pi} d\Theta$$

$$= A + \frac{1}{2\pi} . [\sin(\omega_0 t + \Theta)]_{0}^{2\pi}$$

$$E[X(t)] = A$$

B. Calculate the Autocorrelation.

$$R_{XX}(t_1, t_2) = E[X(t_1).X(t_2)]$$
$$= E[(A + \cos(\omega_0 t_1 + \Theta)\,(A + \cos(\omega_0 t_2 + \Theta))]$$
$$= E[A^2] + E[A\cos(\omega_0 t_1 + \Theta)] + E[A\cos(\omega_0 t_2 + \Theta)] + E[\cos(\omega_0 t_1 + \Theta)\cos(\omega_0 t_2 + \Theta)]$$

$$\boxed{\cos(A) . \cos(B) = \frac{1}{2}(\cos(A + B) + \cos(A - B))}$$

$$= A^2 + 0 + 0 + E\left[\frac{1}{2}(\cos(\omega_0 t_1 + \omega_0 t_2 + 2\Theta) + \cos(\omega_0 t_1 - \omega_0 t_2))\right]$$
$$= A^2 + \frac{1}{2}E[\cos(\omega_0(t_1 + t_2) + 2\Theta)] + \frac{1}{2}E\big[\cos\big(\omega_0(t_1 - t_2)\big)\big]$$
$$= A^2 + 0 + \frac{1}{2}\cos\big(\omega_0(t_1 - t_2)\big)$$
$$= A^2 + \frac{1}{2}\cos\big(\omega_0(t_1 - t_2)\big)$$

C. Calculate the Autocovariance

$$C_{XX}(t_1, t_2) = R_{XX}(t_1, t_2) - E[X(t_1)]E[X(t_2)]$$
$$= A^2 + \frac{1}{2}\cos\big(\omega_0(t_1 - t_2)\big) - A^2$$

$$= \frac{1}{2}\cos\big(\omega_0(t_1 - t_2)\big)$$

3. You are designing a buffer for a video streaming app. The video requires **100 MB** to start playing. Because of a bad Wi-Fi connection, the download progress (measured in Megabytes or MB) behaves randomly. At every second:
   - **Success (50% chance):** You download **1 MB**.
   - **Buffering (50% chance):** You download **0 MB** (stay in place).

   Let $X[n]$ be the total MB received after n seconds.

   A. Your app shows a progress bar to the user. Based on the Ensemble Mean $E\big[X[n]\big]$, after how many seconds should the app tell the user the download will 'likely' be finished?

   B. Using only the ensemble mean to predict the finish time results in a 'premature notification' 50% of the time (the app says it's done, but the file is still buffering). To provide a more reliable estimate, we want to find the time n where even an 'unlucky' user—defined as being one standard deviation ($\sigma_X[n]$) below the mean progress—has finished the download.
   (**Task**: Solve for the time n required to reach the 100 MB target such that $E[X[n]] - \sigma_X[n] = 100$.)

First, let's define the $X[n]$

$$X[n] = \sum_{i=1}^{n} Z[i]$$

where $Z[i]$ is an i.i.d (independent and identically distributed) sequence of step variables with the following PMF (probability mass function):

$$P(Z[i] = 1) = P(Z[i] = 0) = \frac{1}{2}$$

A. Ensemble Mean $E\big[X[n]\big]$

To show when the download will likely to be finished, we need to know the average amount of received MB after n seconds.

$$\begin{aligned}
E\big[X[n]\big] &= E\left[\sum_{i=1}^{n} Z[i]\right] \\
&= \sum_{i=1}^{n} E\big[Z[i]\big] \\
&= \sum_{i=1}^{n} \left(1.\frac{1}{2} + 0.\frac{1}{2}\right) \quad \boxed{Bernoulli\ distribution} \\
&= \frac{1}{2} n
\end{aligned}$$

Since the app need to download 100 MB of data before the video can be played, we need to find value of n so that $E\big[X[n]\big] = 100$.

With $E[X[n]] = \frac{1}{2}n$, we can find that on average, users need to wait for $n = 200$ seconds before the video can be played.

B. To find the standard deviation, we first need to find the variance.

$$Var(X[n]) = E\left[\left(X[n] - E[X[n]]\right)^2\right] = E[X[n]^2] - E[X[n]]^2$$
$$= \left(\sum_{i=1}^{n}\sum_{j=1}^{n} E[Z[i]Z[j]]\right) - \frac{1}{4}n^2$$
$$\boxed{As\ Z[n]\ is\ i.i.d., \quad E[Z[i]Z[j]] = E[Z[i]]E[Z[j]]\ when\ i \neq j}$$
$$= \left(\sum_{i=1}^{n} E[Z[i]^2]\right) + \left(\sum_{i}\sum_{j\neq i} E[Z[i]]E[Z[j]]\right) - \frac{1}{4}n^2$$
$$= n\left(\frac{1}{2}\right) + n(n-1)\left(\frac{1}{2}\right)^2 - \frac{1}{4}n^2$$
$$= \frac{1}{4}(2n + n^2 - n - n^2)$$
$$= \frac{1}{4}n$$

From that, standard deviation can be calculated as $\sigma_X[n] = \sqrt{Var(X[n])} = \frac{1}{2}\sqrt{n}$

Finally, we can find the answer:

$$E[X[n]] - \sigma_X[n] = 100$$
$$\frac{1}{2}n - \frac{1}{2}\sqrt{n} = 100$$
$$n - \sqrt{n} = 200$$
$$n - \sqrt{n} - 200 = 0 \quad \boxed{use\ quadratic\ formula}$$
$$\sqrt{n} = \frac{1 + \sqrt{1 + 800}}{2} \approx \frac{1 + 28.3}{2} \approx 14.65$$
$$n \approx (14.65)^2 \approx 215\ seconds$$

4. You have a noise signal $X(t)$ that represents the background hum of a microphone. The probability density function (PDF) of this noise is **Uniform**, meaning it is equally likely to be any value between **0 and 1** ($f_X(x) = 1 \; for \; 0 \leq x \leq 1$).
   You pass this signal through an amplifier (a memoryless system) that doubles the signal:
   $$Y(t) = 2X(t)$$
   A. What is the new range of the output $Y(t)$?
   B. As the total probability must still equal 1, but the range has changed (from question A), what must happen to the "height" of PDF ($f_Y(y)$)?
   C. Use the transformation formula to find $f_Y(y)$:
   $$f_Y(y) = \frac{f_X(x)}{|J(x)|}$$
   where Jacobian $J(x) = \frac{dy}{dx}$

A. From the definition of:

$$Y(t) = 2X(t)$$

The system behaves like a multiplier. ($Y = 2X$)

- When X is at its minimum (0), $Y = 2(0) = 0$.
- When X is at its maximum (1), $Y = 2(1) = 1$

Therefore, the output Y has range of [0,2].

B. Since the range of output is double the input, the PDF of Y(t) needs to shrink by half, so the total probability is still 1.

C. Firstly we need to find the value of the Jacobian $J(x)$:

$$J(x) = \frac{dy}{dx}$$
$$= \frac{d}{dx}(2x)$$
$$= 2$$

With that, we can find $f_Y(y)$:

$$f_Y(y) = \frac{f_X(x)}{|J(x)|}$$
$$= \frac{1}{|2|}$$

Therefore,

$$f_Y(y) = \frac{1}{2} for\ 0 \leq y \leq 2$$